

3'end RNA-seq using template switching protocol

Status: [?] | partially migrated / unconfirmed

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About

Protocol for 3' end RNA-seq using a template switching reverse transcription workflow.

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1. RNA extraction

Input / starting material: RNA lysate

- Maxwell RNA extraction
- Alternative: Add 3 volumes of LS-Trizol to lysate
- Extract RNA using the Zymo MicroSpin Columns
- Elute with 6 μ L of H₂O

2. RNA fragmentation and template switching RT

- Prepare fragmentation buffer.
- Fragmentation buffer: 80 mM Tris-HCl pH 7.5, 40 mM MgCl₂.

Note: have more recently been using milder fragmentation buffer conditions with input concentration of 10 mM Tris-HCl pH 7.5, 5 mM MgCl₂ (final concentration 1.25 mM Tris-HCl and 0.625 mM MgCl₂).

- Use 500 ng of RNA (whenever possible).

- Bring to 3.5 μ L using water.
 - Add 0.5 μ L of fragmentation buffer (10 mM Tris-HCl pH 7.5, 5 mM MgCl final concentration).
 - Fragment RNA by incubating 12 min at 95 $^{\circ}$ C.
 - Add:
 - 0.5 μ L 10 mM dNTPs
 - 0.5 μ L 5 μ M primer RT1
 - Mix and incubate at 65 $^{\circ}$ C for 3 min then cool down to 42 $^{\circ}$ C and hold.
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3. Template switching RT incubation and cleanup

- Make a mastermix and add 5.5 μ L of the following:
 - component X
 - component Y
 - component Z
 - Place the mastermix into a PCR strip and into the cycler so that it pre-warms to 42 $^{\circ}$ C before you start adding it to the samples.
 - Add mastermix to samples while keeping them at 42 $^{\circ}$ C in the cycler (i.e., don't take the samples out of the cycles while adding the mix).
 - Incubate at 42 $^{\circ}$ C for 10 min.
 - Incubate at 50 $^{\circ}$ C for 30 min.
 - Purify using Magbind beads (1.8 μ L per 1 μ L of PCR reaction). Elute in 8 μ L H₂O.
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4. Optional PCR test

- Use 1 μ L RT product.
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5. Indexing PCR

- Prepare the following 20 μ L indexing PCR reaction for each sample:
 - 10 μ L of 2 $\times$ Q5
 - 2 μ L RT product
 - 1 μ L dual indexed primers
 - * Use a different index for each sample.
 - * Make sure each sample will have a different index if submitting multiple libraries at the same time.
 - 7 μ L of ddH₂O

- Incubate for the PCR reaction using the Q5 program:
 - Extension time – 45 seconds
 - Tm – 65 °C
 - Typically try 10 and 14 cycles
 - Move samples to post-PCR room.
 - Add 2 µL of purple 6X loading buffer to 5 µL of PCR reaction. Keep the rest of the 20 µL.
 - Load on a 6% TBE gel together with 2 µL of ready-to-use low molecular weight ladder (NEB).
 - Run at 180 V for 25 min.
 - Stain with Sybr Green (1:10000 dilution) in 1X TBE.
 - Visualise.
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6. PCR beads purification

- Mix together approx. 10 µL of each PCR reaction.
 - Add a suitable amount of Magbind beads (e.g. 0.9 µL per 1 µL of PCR reaction) and mix.
 - Incubate at room temperature for 5 minutes.
 - Remove supernatant and wash with 500 µL of 70% Ethanol for 30 sec. Repeat this step.
 - Air dry the beads.
 - Resuspend in 30 µL of H₂O.
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7. Buffers

- Fragmentation buffer: 80 mM Tris-HCl pH 7.5, 40 mM MgCl₂.
 - Milder fragmentation buffer conditions: input concentration of 10 mM Tris-HCl pH 7.5, 5 mM MgCl₂ (final concentration 1.25 mM Tris-HCl and 0.625 mM MgCl₂).
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